

Hiya Dullu

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Education

Vellore Institute Of Technology, Bhopal

2023-2027

Bachelor of Technology in Computer Science and Engineering (GPA: 8.42)

Technical Skills

Languages: Python, Java, JavaScript, HTML, CSS

Technologies: : Node.js, Flask, EJS, TensorFlow, OpenCV, MediaPipe, PostgreSQL, MySQL, AWS, Git

Concepts: : Cloud Computing, RESTful APIs, Computer Vision, Machine Learning, Database Management, Version Control, Ubuntu, Windows

Projects

AI-based Cognitive Vision System for Threat Detection | *Python, OpenCV, TensorFlow, YOLOv8, MediaPipe, C++*

- Developed a real-time computer vision system integrating object detection, facial recognition, and hand tracking for automated security threat identification. .
- Deployed YOLOv8 for hazardous object detection, achieving higher inference speed and improved precision compared to YOLOv5 in live video streams, and implemented MediaPipe-based hand gesture analysis to flag anomalous movements using motion thresholds.
- Integrated facial recognition using pre-trained deep learning models to enhance access control and threat assessment, with cross-platform deployment and testing on Ubuntu and Windows for reliable hardware-software interfacing.

QR-Enabled Digital POS and Ledger Management System | *Node.js, Express, PostgreSQL, EJS, Passport.js, Render, Gemini API*

- Architected a modular, web-based restaurant management platform integrating POS systems, QR-based customer ordering, admin dashboards, and automated onboarding workflows.
- Built scalable REST APIs using Node.js and Express with PostgreSQL for transactional data, authentication via Passport.js, and dynamic QR-code-driven customer menus, reducing manual order handling and improving operational efficiency
- Implemented automated deployment notifications and admin-controlled onboarding pipelines, enabling faster restaurant go-live times, with planned AI-driven insights using Gemini API for customer behavior and market trend analysis.

Online Proctored Examination System | *HTML, CSS, JavaScript, Bootstrap, Node.js, Python, WebRTC*

- Built a distributed, web-based examination platform with a Node.js backend handling authentication, timed quiz orchestration, and proctoring event ingestion via asynchronous REST APIs.
- Implemented real-time, asynchronous proctoring using WebRTC for webcam and microphone access, enabling continuous monitoring and event logging under real-world network and device constraints.
- Developed a decoupled Python microservice using PyMuPDF and python-pptx to extract content from instructor-uploaded PDFs/PPTs and generate MCQ candidates, improving system scalability, maintainability, and instructor productivity.

Experience & Achievements

Research Publications

IEEE Conference Paper – Cardiovascular Disease Prognosis Using Ensemble Machine Learning

- Developed a machine learning-based clinical decision support system using the UCI Cleveland Heart Disease dataset with 14 structured clinical attributes including age, chest pain type, cholesterol levels, and ECG results.
- Evaluated Logistic Regression, SVM, Random Forest, XGBoost, and Neural Networks; proposed an ensemble soft-voting model achieving **92% accuracy** and a **macro F1-score of 91%**.
- Presented at an IEEE-affiliated international conference, demonstrating the effectiveness of ensemble learning in structured healthcare analytics.

Software Development Club – General Secretary

- Managed end-to-end logistics and operations for technical events and workshops, coordinating cross-functional teams.
- Developed backend components for the club website, supporting content delivery and internal coordination.

Android Club – Technical Content Lead

- Led technical content strategy for workshops, curating structured materials aligned with Android and backend development practices.

Google Crowdsourcing Hackathon – Operations Manager

- Coordinated and managed **3+ sprint cycles**, overseeing event flow for **50+ participants**.

Muskurahat Foundation – Volunteer

- Led outreach activities during the **April 2024 crowdfunding campaign**.

Technovation Girls (Adobe Mentorship Program) – Student Developer

- Contributed to the development of a **telehealth mobile application** using block-based programming tools.

Certifications

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| • MATLAB Certified – MathWorks | Aug 2023 |
| • Python Pro Bootcamp – Udemy | May 2024 |
| • AWS Certified AI Practitioner – Ethnus | May 2025 |
| • IBM Generative AI – IBM | May 2025 |
| • The Bits and Bytes of Computer Networking – Coursera | Nov 2025 |